⛏ STAR CITIZEN · MINING · GEAR, TECHNIQUE & HOTSPOTS

SC Mining Guide

Equipment in detail and how to use it — plus the best hotspots. Featuring the **Aaron Halo** with exact distances to all **10 bands**.

TYPES	CORE ORE	TOP SPOT	PROSPECTOR BAG
FPS · ROC · Ship	**Quantanium**	**Aaron Halo**	**32 SCU**
PATCH			
4.9			

01 The mining cycle

Applies to hand, ROC and ship.

1 · SCAN

Scan Mode (M)

Ping (Tab), ore rocks light up. Read **Mass** (yield), **Resistance** (power needed), **Instability** (blow-up risk).

2 · FRACTURE

Fracture Mode

Throttle (wheel) into the **green window** and hold until the charge hits 100% → the rock fractures.

3 · EXTRACT

Extraction Mode

Switch, vacuum the valuable chunks, ignore inert ballast. Full (32 SCU) → refinery.

too low	GREEN ▲	yellow	RED →

Laser power (wheel, 0–100%): keep the arrow in the **green window**. Better a bit low & slow than into red = explosion = lost money.

4.9: LASER +20%

All ship mining lasers are ~20% stronger: hard rocks crack easier, but you **overshoot the green window faster** → dose finer, maybe fewer focus modules, more window-wideners.

02 Scan signatures — ID ore before scanning

The number above the rock reveals ore & amount.

In **scan mode**, putting a rock in your crosshair shows a central **signature number** — even **before** you scan. Each resource has a **fixed base value**; the displayed number is **base × number of rocks** in the cluster. So you read ore type **and** quantity at a glance.

HOW TO READ IT

Examples: **3600** = 1× Bexalite · **14,400** = 4× Bexalite (Rare → max 4×) · **6340** = 2× Quantanium (Legendary → max 2×). **Max multiples by rarity:** Legendary 2× · Epic 3× · Rare 4× · Uncommon 5× · Common 6× — no cluster shows more of **one** type than that. Full list in the **print table at the very bottom**. The signature names the **dominant** mineral (40–80%); secondaries (5–20%) show only on a full scan.

TIER	RESOURCE → SIGNATURE (SHIP MINING)
Legendary max 2×	Quantanium **3170** · Stileron **3185** · Savrilium **3200**
Epic max 3×	Ouratite **3370** · Riccite **3385** · Lindinium **3400**
Rare max 4×	Beryl **3540** · Taranite **3555** · Borase **3570** · Gold **3585** · Bexalite **3600**
Uncommon max 5×	Laranite **3825** · Aslarite **3840** · Titanium **3855** · Tungsten **3870** · Agricium **3885** · Torite **3900**
Common max 6×	Hephaestanite **4180** · Tin **4195** · Quartz **4210** · Corundum **4225** · Copper **4240** · Silicon **4255** · Iron **4270** · Aluminium **4285** · Ice **4300**

Ground deposits & salvage

SOURCE	SIGNATURE	NOTE
Small deposit (hand/FPS)	**3000**	100% single mineral (Hadanite, Aphorite, Dolivine, Jaclium, Sadaryx, Carinite …)
Large deposit (ROC/vehicle)	**4000**	100% single mineral
Salvage panels	**2000**	per panel (multiples = more panels)
Wreck debris	**1700–3000**	small 1700 · medium 1850 · large 2400 · capital 3000

IMPORTANT — PATCH-DEPENDENT

These values are as of **SC 4.7** (Diftic signature list). CIG occasionally re-tunes signatures — your **2000** currently matches salvage panels; asteroid minerals sit at 3170–4300. Verify live if unsure (tools below). The mechanism (base × count) stays the same. **Vlk pearls / flowstone** read signature **0** (invisible to scan).

LIVE LOOKUP TOOLS

Enter the number → ore + count: STARVEIN – Scan Signatures · SCMINER – Signature Scanner · Diftic – signature list (GitHub).

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03 Equipment & usage

Prospector (32 SCU, solo) → upgrade ARGO MOLE (3 arms, 96 SCU, crew).

Mining heads (S1)

HEAD	SLOTS	CHARACTER	FOR
Arbor MH1 (default)	3	balanced	all-rounder, good start
Helix I / II	3	high power, narrow window	experienced, hard rocks
Lancet MH1	3	wide window, forgiving	beginners, steady mining
Hofstede S1	2	low power, stable	soft/unstable rocks
Klein S1	1	precise, low spread	small rocks

Passive modules

MODULE	EFFECT
Focus / Focus II	faster fracturing
Rime / Ricol	lowers instability
Brandt	more resistance tolerance
Vaux	lowers minimum power
XTR-L / Optimum	widens the window
Stampede / Torrent	faster extraction
Rieger	+extraction, window −10%
FLTR-XL	−24% inert ballast → more bag space

Active modules & gadgets

MODULE / GADGET	EFFECT (APPROX.)
Sabir (Gadget)	resist. −50%, window +50% → hard rocks
WaveShift (Gadget)	window +100%, instab. −35% → forgiving

MODULE / GADGET	EFFECT (APPROX.)
OptiMax (Gadget)	yield +60%, window −30% → pricey rocks
Stalwart (Gadget)	instab. −35%, rate +50% → fast & stable
BoreMax (Gadget)	instab. −70% → extremely unstable rocks
Okunis (Gadget)	rate +100%, window +50% → speed
Surge (active)	resist. −16%, extr. +50% → stuck rock
Lifeline (active)	instab. −20%, blow-up threshold +60% → emergency brake
Torpid (active)	charge rate +60%, riskier

ONLY ONE GADGET PER ROCK

Two gadgets on the same rock can make it **catastrophically explode**. Active modules (Surge/Lifeline/Torpid) run separately. Values are patch-dependent — the in-game description beats the table if in doubt.

🧩 How to deploy a gadget (step by step)

Gadgets (Sabir, WaveShift, OptiMax, BoreMax, Okunis, Stalwart) are **handheld devices you place directly on the rock** — not into the mining head. The flow:

1. **Buy the gadget** (a consumable) and carry it in your inventory/on your belt.
2. Check the rock in **scan mode** (resistance/instability) → pick the right gadget (see "gadget by problem" below).
3. **Get to the rock:** exit the ship and **EVA** over to the asteroid (for ROC/ground deposits just walk up).
4. **Attach the gadget:** aim at the rock + interact → it sticks and the **gadget UI** opens.
5. **Calibrate:** tune the three dials **propagation**, **field range** and **axial scatter** until you hit the rock's **resonant frequency** (readout peaks).
6. **Activate** → the rock takes on the altered stats (e.g. −resistance, +window) for a **limited time**.
7. Back into the ship / to your mining position and fracture the rock in the **green window**, then extract.
8. After the fracture **pick the gadget back up** (reusable unless destroyed).

TIMING & LIMIT

Only 1 gadget per rock (two = explosion). After activating, **fracture promptly** — the effect wears off.

CALIBRATION — HONEST TAKE

Documented: Three dials — **propagation**, **field range**, **axial scatter** — tuned to the rock's **resonant frequency**; only then can the gadget be activated. The **per-dial effect is not documented by CIG** (detail lives only in video tutorials). **Interpretation (unofficial):** propagation ≈ coarse frequency/spread · field range ≈ reach of the effect field · axial scatter ≈ dispersion/fine alignment. In practice: **adjust the dials until the readout hits resonance/"lock" → activate**. Calibration is fiddly — many miners use the slotted active modules instead; for hard huge rocks the handheld gadgets are the stronger lever. Patch-sensitive.

DIFFERENCE: ACTIVE MODULES

Surge / Lifeline / Torpid work differently: not placed on the rock but slotted into the mining head's **module slots via mobiGlas**, then fired **from your seat** while mining with **Left Shift + 1/2/3** (slot top→bottom) — no EVA needed.

🎯 Gadget by problem

ROCK PROBLEM	FIX
Too hard (resistance too high)	**Sabir** (–50 % Resist.), else Surge
Too unstable (preventive)	**BoreMax** (–70 %) / WaveShift
Going red NOW (emergency)	**Lifeline**
Window too tight / overshoot	**WaveShift** (+100 % window)
Pricey rock, max yield	**OptiMax** (+60 %)
Speed/throughput	**Okunis** / Stalwart

📦 Recommended loadouts

GOAL	HEAD	MODULE
Beginner (forgiving)	Lancet MH1	XTR-L + Rime + Stampede
Advanced (speed)	Helix I	Focus II + Torrent + Surge/Sabir
Quantanium (high-value)	Arbor/Lancet	Rime + XTR-L + Lifeline

TOOLS

Plan rock values & loadouts ahead: Regolith (calc/loadouts), UEX (prices/sell spots), CStone item finder (module values).

MOLE: the 3 laser bars (solo vs. crew)

The three bars are **three independent mining turrets**, each with its own laser and throttle (0.10–100%, its own green window) — **no shared power pool**. There is no "focus into one beam": each laser runs at its **own** max; the others lend it nothing.

MODE	WHAT IT DOES
1 bar only (solo)	Effectively the only solo mode: the **pilot seat can't mine**; you sit in one turret — leave the seat and that laser shuts off. → put your strongest head (e.g. Helix II) on this one.
3 bars parallel (crew)	Three operators on **three different rocks** → up to **3× throughput**.
3 bars concentrated (crew)	All three beams on **the same large/hard rock** → combined power, cracks deposits **>25,000 kg** a single laser can't. Everyone must hold their green window → coordinate.

> NOTE
> "Only one" is unavoidable solo and runs at its own full max. The gain from "all three" is **parallelism** (3 rocks) or **bundling** (1 huge rock) — **not** a stronger single beam.

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04 Aaron Halo — 10 bands with distances

The prospector's home turf: quantanium & taranite.

An asteroid belt ringing all of Stanton — between the orbits of **Crusader and ArcCorp**, reachable from anywhere. It's made of **10 bands** of varying density (CaptSheppard density survey, 1,746 data points, server-stable). **Band 5 is by far the densest** = the jackpot band. Note: bands differ in **density** (how many rocks), **not ore type** — what's inside is per-rock (scan each).

Method A — system-wide (recommended, any route)

The halo sits **equidistant 360°** from the Stanton center marker. Plot **any** quantum route crossing the belt, open the starmap, select the **Stanton system marker** (center) and watch your **distance to the marker** (top right). Abort quantum (hold B) at the target distance → you're in the band you want.

BAND	DISTANCE TO STANTON MARKER (DENSEST POINT)	DENSITY
1	19,702,000 km	very thin
2	19,867,000 km	medium
3	19,995,000 km	solid
4	20,188,000 km	medium
5 ◆	20,320,000 km	**JACKPOT (densest)**
6	20,471,000 km	solid

BAND	DISTANCE TO STANTON MARKER (DENSEST POINT)	DENSITY
7	20,662,000 km	solid (2nd densest)
8	20,881,000 km	solid
9	21,082,000 km	solid
10	21,307,000 km	solid

> TIP
> If you can choose: **Band 5** (≈ 20,320,000 km) — more rocks per ping = less flying around. Solid fallbacks: 7, 3, 6, 10.

📍 Method B — refinery route (example ARC-L1 → CRU-L4)

Plot a fixed destination, start quantum, abort at the **remaining distance to the destination**. Example route **ARC-L1 refinery → CRU-L4**:

BAND	ABORT AT REMAINING DISTANCE TO CRU-L4
1	12,744,803 km
2	13,157,847 km
3	13,510,150 km
4	13,934,165 km
5 ◆	14,292,609 km
6	14,637,246 km
7	15,058,764 km
8	15,524,706 km
9	15,938,061 km
10	16,188,952 km

All other refinery routes (ARC-L1/L2/L4, CRU-L1, HUR-L1/L2, MIC-L1/L2/L5) are in the local PDF `3.19.1-LIVE-Rel1-Refinery to Aaron Halo Mining Routes.pdf` or at CStone (source).

⚡ Quick entry (rough)

- **Hurston → ArcCorp**: exit at ~9,500,000 km remaining.
- **ArcCorp → Hurston**: exit at ~13,500,000 km remaining.
- **Grim Hex → CRU-L2**: stop at 750,000 km.

> **ACCURACY / PATCH**
> The 10-band distances come from CaptSheppard's density survey (densest point per band, to the Stanton marker). The belt structure is server-stable; exact route distances (method B) depend on planet/L-point positions and can drift slightly per patch. Use as a precise start, then fine-tune with the mining scanner.

🔹 Ores in the halo

ORE	TIER	NOTE
Quantanium	TOP	Main source, top value — **unstable, sell immediately** (~150k/SCU).
Taranite	TOP	very common in the halo, top value (~29k).
Borase	TOP	rare, grab on purpose (~33k).
Laranite · Hephaestanite	HIGH	solid, common.
Titanium · Diamond · Gold	MID	solid filler.

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05 Other hotspots

Brief — details in the Prospector PDF.

REGION	FOR	HIGHLIGHT	RISK
Aaron Halo (Stanton)	Ship	Quantanium · Taranite	low–mid (PvP)
Stanton moons	ROC / hand	Hadanite (Aberdeen★/Daymar), Aphorite, Dolivine	low
Pyro	Ship/ROC	Bexalite + Taranite (Pyro I) — lawless, rich	high (pirates)
Nyx (Glaciem Ring)	Ship	Lindinium (~51k) · Bexalite; QV Breaker (Sadaryx)	mid (remote)

> **ARMOR WARNING (ROC/HAND)**
> Hot moons (esp. **Aberdeen**) → heat-resistant armor (Pembroke); cold moons → cold protection. Watch med-pens & O2. **4.9 bug:** standard Greycat ROC broken → use ROC-DS / ROC-GEO.

♻ Refining method, briefly

Expensive ore (quant/bexalite) → high-yield/slow (**Dinyx Solventation**). **Bulk** → fast/cheap (**Cormack**/Kazen). All-rounder: Ferron/XCR. Sell quantanium **immediately** (it decays!).

06 Event minables — location-locked resources

Not farmable "anywhere": one location only, often hand-mining only.

Some ores exist at **one single location** only and are tied to a specific activity — sometimes with no ship/vehicle route at all, i.e. pure **hand-mining**. These are mainly the **Wikelo resources**.

RESOURCE	ONLY HERE	MINING	CATCH
Saldynium	Hathor Caves (Stanton)	hand	nowhere else, **no ship route**
Jaclium	Hathor Caves	hand/laser	tied to the Hathor activity
Carinite	Hathor (special) + regular Aberdeen/Daymar	laser/hand	high density only at Hathor
Sadaryx	QV Breaker Stations (Nyx)	hand (~25%)	only there

> ACCESS = DEDICATED GUIDES
>
> Get these via their activity: **Hathor Caves** (carinite/jaclium/saldynium) → see the Hathor OLP guide ↗; **Sadaryx** → the QV Breaker Stations guide ↗ (reboot station, lenses, laser). Prep: environmental armor + hand/ROC mining tool.

⚠ Hard/risky despite being "everywhere"

Quantanium — highly volatile, decays/explodes under time pressure → refine/sell immediately. **Gem deposits** (hadanite, dolivine, aphorite, janalite, beryl) — only via **FPS/ROC mining** in caves/on the surface, not by ship.

↗ Sources

Values/prices/distances are patch-dependent — verify live.

CStone / CaptSheppard — Aaron Halo Travel Routes & Density Chart (10 bands) ↗

Regolith — mining calculator & loadout planner ↗

UEX Corp — ore prices & best sell locations ↗

Star Citizen Wiki — Mining ↗

Based on: own SC_Prospector_Mining_Guide.pdf (4.9) + CaptSheppard survey. Unofficial, not affiliated with CIG. As of 4.9 (2026). Star Citizen® is a trademark of CIG. All data without warranty.

Print table: scan signatures (with multiples)

Each number = **base × number of rocks** of one type in the cluster. The max count depends on rarity: **Legendary 2× · Epic 3× · Rare 4× · Uncommon 5× · Common 6×**.

MINERAL	TIER	×1	×2	×3	×4	×5	×6
Quantanium	Leg (2×)	3170	6340				
Stileron	Leg (2×)	3185	6370				
Savrilium	Leg (2×)	3200	6400				
Ouratite	Epic (3×)	3370	6740	10110			
Riccite	Epic (3×)	3385	6770	10155			
Lindinium	Epic (3×)	3400	6800	10200			
Beryl	Rare (4×)	3540	7080	10620	14160		
Taranite	Rare (4×)	3555	7110	10665	14220		
Borase	Rare (4×)	3570	7140	10710	14280		
Gold	Rare (4×)	3585	7170	10755	14340		
Bexalite	Rare (4×)	3600	7200	10800	14400		
Laranite	Unc (5×)	3825	7650	11475	15300	19125	
Aslarite	Unc (5×)	3840	7680	11520	15360	19200	
Titanium	Unc (5×)	3855	7710	11565	15420	19275	
Tungsten	Unc (5×)	3870	7740	11610	15480	19350	
Agricium	Unc (5×)	3885	7770	11655	15540	19425	
Torite	Unc (5×)	3900	7800	11700	15600	19500	
Hephaestanite	Com (6×)	4180	8360	12540	16720	20900	25080
Tin	Com (6×)	4195	8390	12585	16780	20975	25170
Quartz	Com (6×)	4210	8420	12630	16840	21050	25260
Corundum	Com (6×)	4225	8450	12675	16900	21125	25350
Copper	Com (6×)	4240	8480	12720	16960	21200	25440
Silicon	Com (6×)	4255	8510	12765	17020	21275	25530
Iron	Com (6×)	4270	8540	12810	17080	21350	25620
Aluminium	Com (6×)	4285	8570	12855	17140	21425	25710
Ice	Com (6×)	4300	8600	12900	17200	21500	25800

Ground / salvage (own rows, more multiples):

ROC mineables (×1–7, base 4000): 4000 · 8000 · 12000 · 16000 · 20000 · 24000 · 28000

FPS/hand mineables (×1–10, base 3000): 3000 · 6000 · 9000 · 12000 · 15000 · 18000 · 21000 · 24000 · 27000 · 30000

Salvage panels (×1–15, base 2000): 2000 · 4000 · 6000 · 8000 · 10000 · 12000 · 14000 · 16000 · 18000 · 20000 · 22000 · 24000 · 26000 · 28000 · 30000

SC 4.7 base values; patch-sensitive. Vlk pearls/flowstone = signature 0 (invisible).